



Naive Bayes Classifier

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SUBMISSION DATE: 01.11.2025

Introduction

The objective of this lab is to explore probabilistic text classification using the Naive Bayes algorithm, a machine learning approach for handling discrete features such as word frequencies.

There are 3 tasks performed:

- **Part A:** Implementation of the Multinomial Naive Bayes classifier using by computing log priors and log likelihoods, with the application of Laplace smoothing to address zero probabilities.
- **Part B:** Application of Scikit-learn's MultinomialNB model with a TF-IDF vectorizer pipeline, followed by hyperparameter tuning using GridSearchCV to identify optimal model settings.
- **Part C:** Approximation of the Bayes Optimal Classifier (BOC) using an ensemble of diverse models - MultinomialNB, Logistic Regression, Random Forest, Decision Tree, and K-Nearest Neighbors. These models are combined using a Soft Voting Classifier, with posterior weighting applied to each model's contribution.

Methodology

Part - A

1. **Data Preprocessing:** Sentences and corresponding section labels were loaded from the PubMed RCT dataset. Text data was converted into numerical feature vectors using CountVectorizer, where each feature represented the frequency of a word in a sentence.
2. **Model Training:** The log prior probabilities for each class were computed based on their relative frequencies. The log likelihoods for each word given a class were calculated with Laplace smoothing to avoid zero probabilities.
3. **Prediction:** For each sentence, the model summed the log prior and log likelihood values to obtain the total log probability for each class. The class with the highest log probability was selected as the prediction.
4. **Evaluation:** The trained model was evaluated on the test data using accuracy, macro F1-score, and a confusion matrix heatmap to visualize performance.

Part - B

A Pipeline combining a TF-IDF vectorizer and a Multinomial Naive Bayes classifier was optimized using GridSearchCV, testing different n-gram ranges and smoothing values.

The tuning used 3-fold cross-validation with f1 macro scoring to select the best-performing configuration.

Part - C

The aim here is to approximate the Bayes Optimal Classifier using a Soft Voting Ensemble that combined Multinomial Naive Bayes, Logistic Regression, Random Forest, Decision Tree, and K-Nearest Neighbours.

1. Sampling:

A dynamic sample size was determined using the SRN-based formula to ensure the training

data represented the overall distribution.

2. Posterior Weight Estimation:

The training set was split into smaller training and validation portions. Each model's validation log-likelihood was calculated and normalized to derive its posterior weight.

3. Model Refitting:

All five models were retrained on the full sampled training data to finalize their parameters.

4. Soft Voting Ensemble:

The models were combined into a soft voting ensemble, where each model's contribution was weighted according to its posterior estimate.

5. Evaluation:

The ensemble's performance was assessed on the complete test set using accuracy, macro F1-score, and the confusion matrix.

Results and Analysis

Part - A

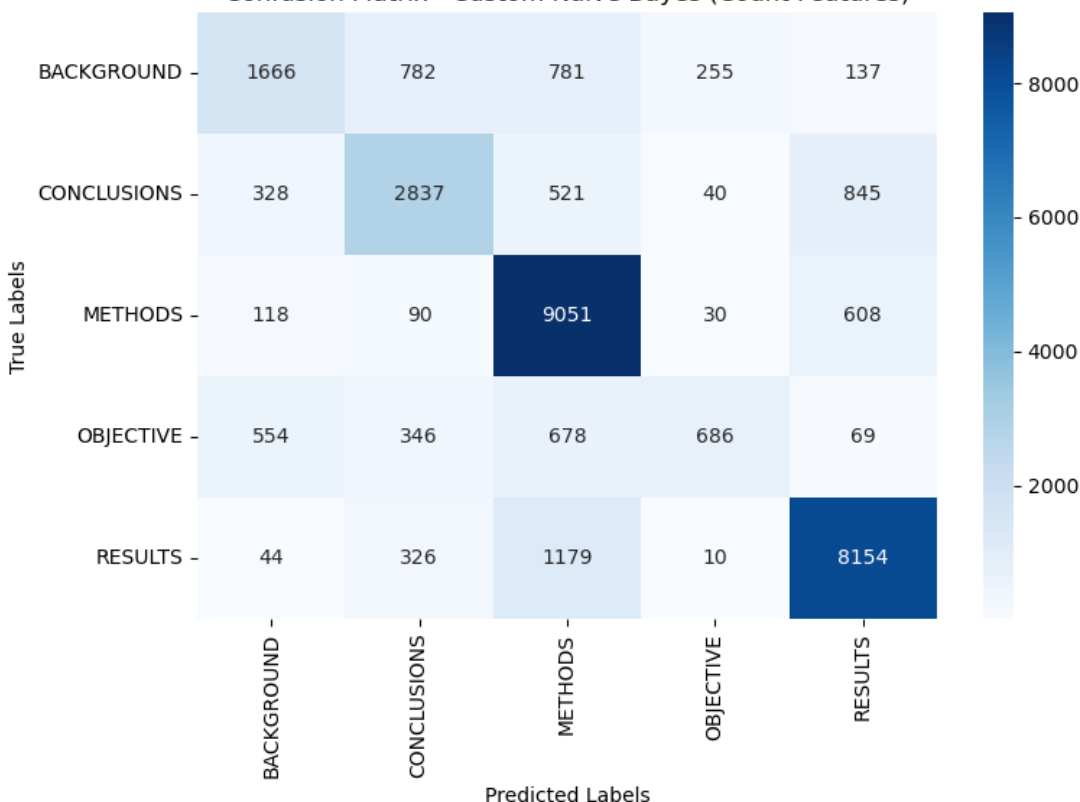
=== Test Set Evaluation (Custom Count-Based Naive Bayes) ===

Accuracy: 0.7431

	precision	recall	f1-score	support
BACKGROUND	0.61	0.46	0.53	3621
CONCLUSIONS	0.65	0.62	0.63	4571
METHODS	0.74	0.91	0.82	9897
OBJECTIVE	0.67	0.29	0.41	2333
RESULTS	0.83	0.84	0.84	9713
accuracy			0.74	30135
macro avg	0.70	0.63	0.64	30135
weighted avg	0.74	0.74	0.73	30135

Macro-averaged F1 score: 0.6446

Confusion Matrix - Custom Naive Bayes (Count Features)



Part - B

Best Hyperparameters:

```
{'nb__alpha': 0.1, 'tfidf__min_df': 3, 'tfidf__ngram_range': (1, 3)}
```

Best Cross-Validation F1 (macro): 0.7044

Part - C

Please enter your full SRN (e.g., PES1UG22CS345): PES2UG23CS365

Using dynamic sample size: 10365

Actual sampled training set size used: 10365

=== Final Evaluation: Bayes Optimal Classifier (Soft Voting) ===

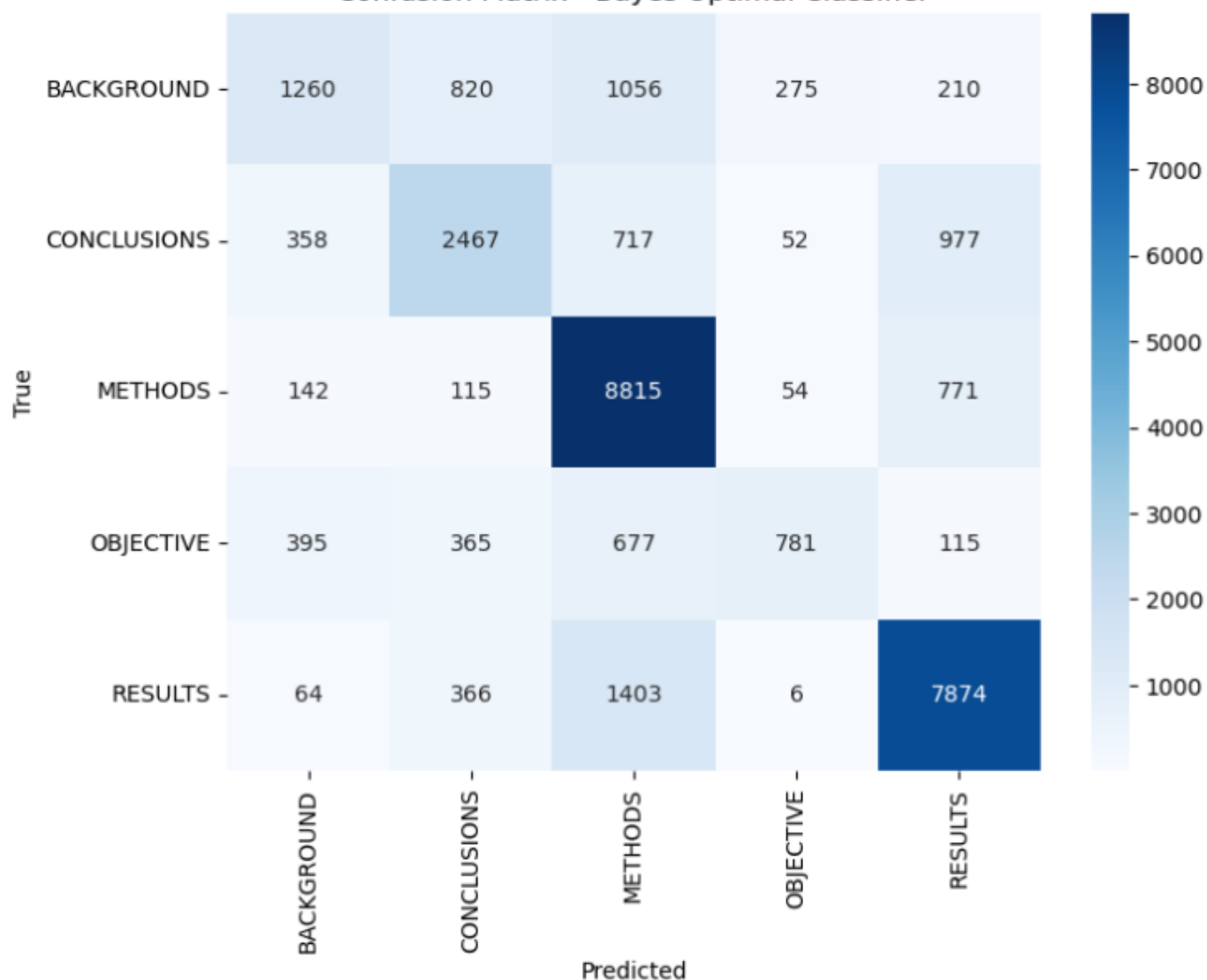
Accuracy: 0.7034

Macro F1 Score: 0.6054

Classification Report:

	precision	recall	f1-score	support
BACKGROUND	0.57	0.35	0.43	3621
CONCLUSIONS	0.60	0.54	0.57	4571
METHODS	0.70	0.89	0.78	9897
OBJECTIVE	0.67	0.33	0.45	2333
RESULTS	0.79	0.81	0.80	9713
accuracy			0.70	30135
macro avg	0.66	0.58	0.61	30135
weighted avg	0.69	0.70	0.69	30135

Confusion Matrix - Bayes Optimal Classifier



Discussion

Part - A: Despite being simple, the model captured the probabilistic nature of text classification effectively. The METHODS and RESULTS classes showed strong recall and F1 scores, while OBJECTIVE and BACKGROUND were relatively weaker, likely due to overlapping vocabulary and limited context captured by unigram features. This baseline performance reflects the effectiveness of the Multinomial Naive Bayes assumption, though it lacks advanced text representation and parameter tuning.

Part - B: After tuning the sklearn-based MultinomialNB model, the performance improved noticeably. Using TF-IDF features and optimizing the hyperparameters helped the model capture more context and finer differences in word usage. This led to a more robust and generalizable classifier, with better F1-scores compared to the initial implementation.

Part - C: The Bayes Optimal Classifier approximation combined five different models through soft voting with posterior weights. While ensembles usually boost stability, this one performed slightly below the tuned Naive Bayes model. The difference likely came from how the classifiers behaved individually and the limited precision of the posterior weighting. Still, the approach worked well to show how multiple models can be combined to approximate the theoretical Bayes optimal decision boundary.